Mend-Amar Badral

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Driven by a problem solving mindset and fueled by curiosity, I aspire to develop genuine solutions for the real-world, impactful problems.

Skills

Languages:	English (C1 IELTS 8.0)
Programming:	Python, Julia, MATLAB, C/C++, R, SQL
Tools:	Linux, Git, Bash, PyTorch, Docker, Quarto, LaTeX

Education

Budapest University of Technology and Economics (BME) — Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering — Budapest, Hungary
Supervisor: Márton Vaitkus
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) — Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics — Ulaanbaatar, Mongolia
GPA: 3.2/4.0
Supervisor: Galtbayar Artbazar
Thesis: **Evaluating land degradation using image processing methods**

Experience

Center of Mathematics Application, NUM — Oct, 2023 – June, 2025
Undergraduate Research Intern

- Developed a custom **Python command line scripts** for processing a total of 1800 multispectral drone imagery dataset capture collected from 5 field areas and calculating vegetation indices and K-means segmentation maps for the vegetation.
- Architected a unified, documented data storage structure for research backups, directly improving data retrieval efficiency and establishing a data management guidelines for future students.
- Containerized data processing pipelines by deploying OpenDroneMap within Docker applications; configured environment dependencies and tuned parameters to ensure reproducible execution across CLI-driven Linux environments.
- Engaged in field survey data collection and multispectral measurements; gained first-hand insight into ecological data gathering to better design the drone imagery analysis with domain-specific research needs.

Projects

aiSim LiDAR point cloud sensor fusion for 3D Gaussian Splatting

- Developed a Python scripts to colorize LiDAR point cloud measurements obtained from aiMotive's aiSim vehicle simulation platform to train 3D Gaussian Splatting in nerfstudio.
- Engineered the transformation between 3D world to camera sensor coordinate frame, made the observations into a consistent coordinate convention by keeping track of each 3D data processing softwares.

Deep Learning Foundations and Concepts Book Examples Reproduction

- Reproduced machine learning demonstrations from Bishop & Bishop's *Deep Learning Foundations and Concepts* using Julia notebooks and utilized innovative interactive sliders with reactivite programming environment using the Pluto.jl package.

Awards

Stipendium Hungaricum Scholarship — Sep, 2025